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Tampering in Enterprise Monitoring Systems

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Chapter 1

1.0

Introduction

Digital evidence is a crucial component for businesses across all sectors, helping them to investigate security incidents, pinpoint malicious actors and adhere to statutory obligations. The integrity of this evidence: log files, file system metadata, timestamps and audit trails, forms the basis of the entire discipline of digital forensics and is a critical assumption of enterprise security monitoring. However, this assumption is now fundamentally challenged by the advent of agentic artificial intelligence (AI), defined as autonomous systems that can plan and execute multi-stage operations without continuous human oversight (Acharya, Kuppan and Divya, 2025).

In a previous paper I submitted to the Artificial Intelligence Safety and Security Conference (AISS 2026), It showed that an autonomous AI agent can carry out a full anti-forensic operation; destroying evidence, altering timestamps and wiping logs in under 300 milliseconds. This is a deeply unsettling outcome for forensic practitioners: agentic AI, with its transformative potential across industries, can equally be weaponised to undermine the very evidence that investigators rely upon. However, despite this growing threat, there are surprisingly few tools or frameworks available to counter the harm autonomous AI agents can cause to digital forensic processes. The gap between the offensive capabilities of agentic AI and the defensive tools available to forensic experts is rapidly widening. This realisation, that the tools to fight back are not enough, is what drives me, after researching the extent to which agentic AI can evade traditional forensic monitoring, to pivot my focus to building practical countermeasures. This dissertation is thus an immediate response: designing and evaluating mitigation tools to help forensic experts and security teams detect and defend themselves against the very threats exposed in my previous study.

Industry-standard SIEM systems like Wazuh provide File Integrity Monitoring (FIM) through periodic polling, or checking file system states at fixed intervals. This architecture creates a time-of-check to time-of-use (TOCTOU) vulnerability that high-speed autonomous adversaries can exploit. No published study has systematically designed and evaluated defensive countermeasures to detect and mitigate autonomous anti-forensic activity at machine speed, even though this vulnerability has been empirically shown to be exploitable.

1.1

Aim and Objectives

Aim

To design and empirically evaluate an integrated countermeasure framework capable of detecting and mitigating autonomous AI-driven anti-forensic operations that evade standard enterprise monitoring configurations.

Objective

1. To conduct a critical review of existing literature on anti-forensic techniques, TOCTOU vulnerabilities, and agentic AI in offensive security, identifying at least one clearly defined gap in current defensive capabilities, to be completed by the end of Week 2 (1st June 2026).
2. To design and implement an integrated countermeasure framework comprising five modules (real-time event-driven monitoring, behavioural anomaly detection, timestamp validation, canary-based deception, and log integrity protection) with all modules functionally tested by the end of Week 4 (15th June 2026).
3. To evaluate the countermeasure framework against a validated autonomous anti-forensic agent across a minimum of 30 controlled experimental trials, collecting quantitative measurements of detection latency (in milliseconds), detection completeness (percentage of attack stages detected), false-positive rate, and evidence preservation, to be completed by the end of Week 8 (13th July 2026).
4. To statistically compare the detection performance of the countermeasure framework against default enterprise SIEM configurations (periodic FIM) using inferential statistical tests (Mann-Whitney U) at the $p < 0.05$ significance level, with analysis completed by the end of Week 10 (27th July 2026).
5. To produce evidence-based recommendations for enterprise security configurations, derived directly from the experimental results, and present findings within the completed dissertation by the submission deadline of 18th August 2026.

1.2

Research Questions

The proposed research is guided by the following questions:

1. To what extent can real-time, event-driven monitoring detect autonomous anti-forensic operations that completely evade periodic File Integrity Monitoring?
2. Is it possible to apply techniques from behavioural analysis to detect patterns characteristic of agentic anti-forensic execution, for example operation bursts and rapid file lifecycle events, and to differentiate these from normal system activity?
3. How effective are canary-based deception mechanisms at detecting and attributing autonomous anti-forensic operations?
4. What is the detection latency, detection completeness, and false-positive rate of an integrated countermeasure framework compared to default enterprise SIEM configurations when subjected to autonomous anti-forensic attack?

Hypothesis - An integrated countermeasure framework combining kernel-level real-time monitoring, behavioural anomaly detection, timestamp validation, canary-based deception, and log integrity protection will achieve statistically significant improvements in detection rate and forensic evidence preservation compared to standard periodic FIM configurations when subjected to autonomous anti-forensic attack at sub-second execution speed.

1.3

Distinction from Prior Work

It is important to clarify how this dissertation differs from my prior accepted paper (AISS 2026). The two studies address complementary but distinct research questions:

Table 1

Aspect	Prior Paper (AISS 2026)	This Dissertation
Focus	Offensive - demonstrating the attack	Defensive - designing and evaluating countermeasures
Research Question	Can an autonomous agent evade enterprise FIM?	How do we detect and mitigate such agents?
Contribution	Identified the TOCTOU Time Gap vulnerability	Proposes and validates a framework to close it
Artefact	Anti-Gravity attack agent	AntiGravity Shield defence framework
Evaluation	Single-condition (attack vs default SIEM)	Comparative (default SIEM vs integrated countermeasures)
Statistical Rigour	Descriptive (reviewer-noted limitation)	Inferential (Mann-Whitney U, 30+ trials)
Scope	Attack-only	Attack + Defence + Human-baseline comparison

The prior paper established the problem, this dissertation proposes and evaluates the solution.

1.4

Expected Results

It is anticipated that the integrated countermeasure framework will demonstrate:

- Near-complete detection coverage across all three anti-forensic stages, significantly outperforming the zero-detection outcome observed under default periodic FIM in the prior study.
- Sub-second detection latency through event-driven kernel monitoring, effectively closing the TOCTOU Time Gap.
- Low false-positive rates due to behavioural thresholds calibrated to distinguish automated machine-speed operations from normal user activity.
- Enhanced forensic evidence preservation through hash-chained immutable logging and canary-based attribution.

The research will contribute empirical evidence to support recommendations for enterprise security configurations against agentic AI-driven threats and advance the academic understanding of the offensive-defensive balance in the context of autonomous AI systems.

Chapter 2

Literature Review

This chapter provides a systematic examination of the academic and practitioner literature relevant to this research. It is structured thematically, progressing from the foundational concepts of anti-forensic techniques and enterprise monitoring architectures, through the emerging threat of agentic AI in offensive security, to the defensive countermeasures that form the basis of the proposed framework. The chapter concludes by synthesising the literature to identify the specific research gap this dissertation addresses.

2.1 Anti-Forensic Techniques

2.1.1 Anti-Forensic Taxonomy

The academic study of anti-forensics, techniques purposely developed to obstruct or confuse the digital forensic process, originates from the work of Garfinkel (2007), who created the basic taxonomy consisting of three main classes: evidence destruction, evidence hiding, and elimination of evidence sources. This classification was later refined and expanded by later researchers. More granular categorisation was suggested by Rogers (2006) distinguishing data hiding, artefact wiping, trail obfuscation and attacks against the forensic tools themselves. More recently, Conlan, Baggili and Breitingner (2016) carried out a systematic review of anti-forensic techniques, noting that the increasing sophistication of adversary tradecraft requires taxonomies that are constantly updated.

Sabillon and Bermeo (2024) published the most comprehensive modern classification of anti-forensic methods in desktop, mobile, cloud, and IoT environments. This shows that advanced anti-forensic toolkits are integrating timestamp manipulation, secure deletion, and log sanitation into coordinated workflows, which applies to this dissertation's agentic automation.

2.1.2 Timestomping

Timestomping is the intentional modification of file system metadata, in particular the Modified, Accessed, Created, and Entry Modified (MACE) timestamps, to hide the actual timeline of file activity (MITRE, 2024). In the MITRE ATT&CK framework, this technique is classified as T1070.006 under the 'Indicator Removal' tactic, as it is a defence evasion method.

Carvey (2014) examined the manipulation of timestamps in NTFS file systems in detail and showed how the Windows API `SetFileTime` can be used to change the timestamps in the `$STANDARD_INFORMATION` without erasing forensic evidence left in the `$FILE_NAME` attribute and the change journal `$UsnJrnl`. Kroll (2025) analyses this for Linux file systems

(Ext3, Ext4 and FAT32) and demonstrates how `utimensat()` system calls can modify both `mtime` and `atime` fields, while `ctime` (change time) is automatically updated by the kernel, which can be a detection vector. This difference between `ctime` and `mtime` where a file's content modification time is earlier than its metadata change time is a major heuristic exploited by the Timestamp Validator module in the countermeasure framework of this dissertation.

2.1.3 Data Wiping and Log Tampering

In a seminal paper, "Remembrance of Data Passed" (Garfinkel and Shelat 2003), they showed that standard file deletion only removes directory entries, leaving recoverable data on the storage medium. Secure deletion techniques mitigate this by overwriting the contents of the file before it is unlinked. The U.S. Department of Defence standard DoD 5220.22-M requires at a minimum a three-pass overwrite protocol; random data, the complement and another random pass, for sanitisation (NIST, 2014).

Log manipulation is the third pillar of anti-forensic operations. System and application logs are important sources of evidence for forensic timeline reconstruction. Attackers can truncate, overwrite or selectively edit log files to erase traces of their activity (Kent and Souppaya, 2006). Log integrity is particularly vulnerable in enterprise environments if monitoring agents lack real-time protection mechanisms, given that log files can be modified or deleted between the time they are written and the time they are collected by the SIEM (Chuvakin, Schmidt and Phillips, 2013).

2.2 Enterprise Monitoring Architectures

2.2.1 SIEM Systems and File Integrity Monitoring

Security Information and Event Management (SIEM) platforms are the central nervous system of enterprise security operations, aggregating, correlating and alerting on security-relevant events from across the IT estate (Chuvakin, Schmidt and Phillips, 2013).

File Integrity Monitoring is an important capability in modern SIEM deployments, monitoring for unauthorised changes to critical system files, application binaries and configuration data. The leading open-source SIEM platform Wazuh implements FIM in a periodic auditing model. This process is described explicitly in the Wazuh documentation. The FIM module "runs periodic scans... comparing the current checksums of monitored files against previously stored values" (Wazuh Documentation Team, 2024). Default settings are 12 hours between scans (a `

syscheck` frequency of 43200 seconds) . Shorter intervals may be set by administrators, at some cost to computational overhead.

Commercial SIEM platforms share similar architectural patterns. Splunk's Universal Forwarder for log collection, configurable checkpointInterval settings IBM QRadar and Microsoft Sentinel's periodic baseline comparison models. Although NIST SP 800-137 promotes continuous monitoring as the ideal information security assurance state, it recognises that resource constraints such as cost, human resources and technical complexity often lead to organisations adopting periodic assessment configurations (Dempsey et al., 2011). This pragmatic compromise brings the vulnerability that is exploited in this research.

2.2.2 Polling versus Event-Driven Models

This research is based on the distinction between periodic polling and event-driven monitoring architectures. In a polling model the monitoring system only polls at fixed intervals. This means that any activity that happens in between polls will not be seen. In an event-driven paradigm, the operating system's kernel informs the monitoring application in real time about file system events.

On Linux systems, this event-driven capability is provided by the `inotify` kernel subsystem. `inotify` was introduced in kernel version 2.6.13 and operates on a callback mechanism which allows user-space applications to be informed immediately when specified events (creation, modification, deletion, attribute changes) happen to monitored files or directories (Love, 2005). Unlike polling, which uses CPU cycles proportional to the number of files being watched, `inotify` uses almost no resources when idle, firing only when events actually happen. However, `inotify` is known to have limitations that matter to security applications: it does not support recursive watch out of the box (it will need explicit watches on all subdirectories), it is limited by the system-wide number of watches and it does not provide process identification for the actor responsible for the event (Kerrisk, 2010).

2.3 TOCTOU Vulnerabilities in Security Monitoring

The well-characterised class of vulnerability known as the Time-of-Check to Time-of-Use (TOCTOU) race condition occurs when a program checks a condition and then performs an action based upon that condition, but the state of the system changes between the check and the action (Bishop and Dilger, 1996). Wei and Pu (2005) provide a basic anatomical study of TOCTOU vulnerabilities in Unix-like file systems, demonstrating how the non-atomic nature of check-then-use sequences can be abused by concurrent processes.

While the original TOCTOU literature discusses privilege escalation via symbolic link attacks, this work finds a structural analogy in enterprise monitoring architectures: periodic FIM is a "check" operation and the next polling cycle is the next "use" of the same state information. So anything that happens and ends between the two events is in a temporal blind spot. The main difference is scale: While traditional TOCTOU attacks rely on race windows of the order of microseconds, monitoring TOCTOU attacks span seconds to hours and are therefore trivially exploitable by any sufficiently fast adversary.

Tsafrir, Hertz and Wagner (2008) show that even small race windows can be exploited with high probability with repeated attempts. In the monitoring context, the "race" is won in every attempt because the gap between polling cycles is exponentially longer than the time needed for a

machine speed anti-forensic operation. This is confirmed empirically in (Orji et al., 2026), where the autonomous agent completed all anti-forensic stages in 370 milliseconds against a 5-second polling interval.

2.4 Agentic AI in Offensive Cyber Security

2.4.1 Autonomous Agent Architectures

Agentic AI, which are self-driving systems that can understand their surroundings, think about their objectives, and carry out multi-step plans without constant human guidance, is a big change in how AI is used for both good and bad things (Acharya, Kuppan, and Divya, 2025). Agentic systems differ from traditional automation scripts, which follow pre-defined execution paths, in that they are able to adapt their behaviour based on feedback from the environment, making them fundamentally more capable and more unpredictable.

Fang et al. (2024) demonstrated that large language model (LLM) agents can autonomously exploit real-world vulnerabilities given CVE descriptions, achieving success rates that highlight the offensive potential of the technology. Their study suggests that the threshold for autonomous cyber operations has been substantially lowered due to the emergence of general-purpose reasoning models.

2.4.2 AI-Driven Attack Frameworks

Challita and Parrend (2025) introduce RedTeamLLM, an agentic framework for offensive security that implements the “Observe, Plan, Act” loop to perform autonomous reconnaissance, exploitation, and post-exploitation activities. Likewise, Li, Chen and Xu (2023) demonstrate the automation of multi-stage cyber operations using AI, detailing how autonomous agents can link discrete attack techniques into full operational workflows.

The ARTEMIS study (Lin et al., 2025) offers empirical evidence that autonomous AI agents can be competitive with certified human penetration testers. In a comparative evaluation on a large operational university network of ca. 8,000 hosts, the ARTEMIS multi-agent framework came in second place overall, finding nine valid vulnerabilities with an 82% valid submission rate, outperforming nine out of ten OSCP-certified professionals. The cost savings are significant: ARTEMIS costs \$18 per hour compared to \$60 per hour for human testers. In their survey on the AI-based cyber threat landscape, Kaloudi and Li (2020) analyse the ability of AI-powered malware to adapt its behaviour to evade detection. When applied to anti-forensic operations, this capability gives rise to the class of threat that this dissertation addresses.

The operational distinction between manual and agentic anti-forensic activity is vital. Manual attacks are limited by human reaction time (typically hundreds of milliseconds for each individual operation), leave residual artefacts during the decision-making process and are detectable by monitoring systems tuned to human speed activity. In contrast, agentic systems run complete anti-forensic procedures in sub-second time frames, taking advantage of the time blind spots in periodic monitoring architectures.

2.5 Defensive Countermeasures

2.5.1 Real-Time Monitoring Approaches

A variety of researchers and practitioner bodies have promoted the move from periodic to event-driven monitoring. The Center for Internet Security (CIS) recommends real-time FIM for critical system files (CIS, 2023), and industry analyses by CrowdStrike (2024) and Cimcor (2024) demonstrate the increasing awareness that polling-based FIM introduces unacceptable detection gaps in today's threat environments.

Rx-Int (Juneja, 2025) is a kernel-level engine for real-time detection and analysis of in-memory threats, demonstrating the feasibility of kernel-event-driven detection architectures that operate at machine speed. This work validates the architectural principle behind the Real-Time File Monitor module of this dissertation: kernel-level event notification can achieve detection latencies orders of magnitude lower than periodic polling.

2.5.2 Deception-Based Defence

Deception technology, and the utilisation of honeypot or “canary” files specifically, as a detection technique has attracted a significant amount of academic interest. Spitzner (2003) provided the conceptual foundation for honeypot-based intrusion detection. More recent efforts have focused on the detection of ransomware specifically through canary file monitoring. Strategic placement of decoy files can act as high-fidelity tripwires, as Moore (2016) showed, since any access, modification or deletion is a strong indicator of malicious activity, as legitimate processes have no reason to interact with these files.

Al-Rimy, Maarof and Shaïd (2018) reviewed the ransomware countermeasures methodically and noted that canary-based detection is particularly effective due to its low false positive rate, a factor that sets it apart from heuristic and behavioural methods. Further research by Genç, Lenzini and Ryan (2018) demonstrates that canary files can lower the Mean Time To Detect (MTTD) to almost zero, offering an early warning layer that augments signature-based and behavioural detection methods.

2.5.3 Behavioural Anomaly Detection

The detection of behavioural anomalies in file system activity has progressed from simple threshold-based alerting to advanced pattern recognition. Garcia-Teodoro et al. (2009) present an important survey of anomaly-based network intrusion detection, upon which the concept that deviations from established behavioural baselines may be indicative of malicious activity is based. The same principle applies naturally to file system monitoring where the temporal characteristics of operations (frequency, sequencing and velocity) are diagnostic.

Behavioural patterns are highly distinctive in the context of anti-forensic operations. The pattern of "modify, rename, delete" typical of secure file wiping, the rapid sequence of operations when automation is involved, and the creation and subsequent destruction of temporary files all create behavioural signatures that are markedly different from normal user or system activity. In the behavioural detection module of this dissertation, these temporal patterns are used in a sliding-window analysis and a rule-based approach is presented as an alternative to the machine-learning-heavy approaches from the recent literature.

2.5.4 Tamper-Proof Logging

Audit log integrity is a prerequisite for forensic trustworthiness. If an attacker can change or delete logs after the fact, the entire record of evidence is compromised. Hash-chained logging, in which each log entry includes the cryptographic hash of the previous entry, making a mathematical dependency chain, gives a clear record of what was changed (Schneier and Kelsey, 1999). Changing any part of a historical record invalidates all of the subsequent hashes, so you can see if it has been changed.

The blockchain-based logging frameworks are an evolution of hash-chained approaches, which bind cryptographic proofs to distributed ledgers for non-repudiation (Ahmad et al., 2023). However, the performance cost of blockchain-based systems makes them impractical for sub-second detection requirements of this research. The Log Protector module in this dissertation implements a lightweight hash-chain style approach thus delivering tamper evidence without the latency penalties of distributed consensus mechanisms

2.6 Research Gap and Contribution

The literature reviewed in this chapter has shown a clear and exploitable asymmetry: the offensive capabilities of agentic AI in anti-forensic contexts are documented and have been empirically demonstrated, but no published study has systematically designed, implemented and evaluated an integrated defensive countermeasure framework against these threats.

In particular, the literature gap exists at the intersection of three domains:

1. Anti-forensic automation: While individual anti-forensic techniques are well documented (Garfinkel, 2007; Carvey, 2014; Kroll, 2025), their autonomous, coordinated execution by agentic AI systems is addressed (Orji et al., 2026)
2. Monitoring architecture vulnerability: The TOCTOU vulnerability in periodic FIM is structurally analogous to the classic race condition (Wei and Pu, 2005), but no study has proposed countermeasures specifically designed to close this temporal blind spot against machine-speed adversaries.
3. Integrated defensive frameworks: The literature contains individual countermeasure techniques – real-time monitoring, deception, behavioural analysis and tamper-proof logging – in isolation, but no published work has combined these into a single integrated framework and empirically evaluated it against an autonomous anti-forensic agent.

This dissertation directly addresses this gap by designing, implementing, and evaluating the AntiGravity Shield, an integrated countermeasure framework that consists of five coordinated modules, against a validated autonomous anti-forensic agent within a controlled experimental environment

Chapter 3

Methodology

This chapter presents the research methodology adopted for this dissertation, including the philosophical underpinning, the selected research framework, the experimental design, data collection procedures, statistical analysis plan, and ethical considerations. Each methodological decision is justified with reference to the academic literature and the specific requirements of the research questions.

3.1 Research Philosophy and Approach

The approach of this research is a positivist one and the objective is to establish objective and measurable facts about the performance of defensive countermeasures against autonomous anti-forensic operations. The positivist approach is appropriate since the research questions require quantitative answers, detection latencies in milliseconds, detection completeness in percentages and false-positive rates confirmed by statistical hypothesis testing (Saunders, Lewis and Thornhill, 2019). The research is deductive; a hypothesis is formulated from the literature and previous empirical findings, an artefact is constructed to test it and experimental data are collected to confirm or refute it.

The research approach is generally applied and not only theoretical. The dissertation is not intended for developing new theory, but rather designing, building and empirically evaluating a practical artefact, the AntiGravity Shield countermeasure framework, which is intended to solve a real-world problem that has been identified. This applied focus is aligned with the Design Science Research Methodology outlined in the next section.

3.2 Design Science Research Methodology (DSRM)

3.2.1 Justification for DSRM

This research makes use of the Design Science Research Methodology (DSRM) as formalised by Peffers et al (2007). DSRM is a well-accepted research methodology for information systems and computer science research. It is the practice of creating and systematically evaluating IT artefacts such as software tools, frameworks, or models to solve identified problems. It is characterised by its focus on building and evaluation, a natural choice for research where the primary contribution is a functional software framework, rather than purely empirical or interpretivist methodologies.

The foundations of design science in information systems are formalised by Hevner et al. (2004) who suggest that design science research should produce a “viable artefact in the form of a construct, a model, a method, or an instantiation” and should be rigorously evaluated against the desired outcome. This dissertation fulfils both requirements. The artefact is the AntiGravity Shield framework (a software instantiation) and the evaluation is carried out by controlled experimental trials with quantitative performance metrics.

Another methodology considered was a purely experimental approach, without the DSRM framing. However, this was dismissed as the research involves the construction and evaluation of a novel artefact, both of which DSRM explicitly supports in a single coherent framework. A purely experimental approach would not have the structured design rationale that DSRM provides.

3.2.2 Application of DSRM Phases

The six phases of the DSRM process model are applied to this research as follows:

Phase 1 - Problem Identification and Motivation: The TOCTOU Time Gap vulnerability in enterprise periodic FIM was identified and empirically demonstrated in the author's previous research (Orji et al., 2026). The autonomous Anti-Gravity agent completed all anti-forensic operations in an average of 370 milliseconds, and the SIEM configured with a 5 second polling interval, reported zero alerts. This phase has been completed and forms the empirical basis for the present work.

Phase 2 - Definition of Objectives: The countermeasure framework has quantitative objectives: a detection latency of less than one second, detection of at least 75% of the attack stages, zero false-positive rate during quiescent operation, and maintenance of forensic evidence through tamper-evident logging. The objectives are directly based on the research questions (Chapter 1, Section 1.5).

Phase 3 - Design and Development: The AntiGravity Shield framework was designed as an integrated system consisting of five coordinated modules: a real-time file monitor, a timestamp validator, a behavioural pattern detector, a canary deployer, and a log protector. Chapter 4 describes the architectural rationale and the implementation details.

Phase 4 - Demonstration: The framework is deployed in a controlled experimental environment and attacked by the validated Anti-Gravity autonomous agent. The test harness automates the whole experimental workflow enabling repeatable consistent trial execution.

Phase 5 - Evaluation: Quantitative performance data are obtained from at least 30 independent trials per condition and analysed using inferential statistical methods (see Section 3.5). This phase addresses the AI-SS 2026 reviewers' recommendation for greater statistical rigour.

Phase 6 - Communication: Results are communicated in this dissertation and may feed into the follow-on publication for the AI-SS 2026 work.

3.3 Experimental Design

3.3.1 Two-Condition Comparative Design

The study employs a two-condition comparative experimental design to evaluate the effectiveness of the proposed countermeasure framework relative to the current industry-standard baseline:

Control Condition – Enterprise monitoring with default periodic File Integrity Monitoring. In this case, the Wazuh SIEM is configured with the default `syscheck` frequency of 43,200 seconds (12 hours), with the default deployment configuration as documented by the Wazuh Documentation Team (2024). As shown in the previous study, the autonomous agent performs all anti-forensic operations within sub-second timeframes, making periodic FIM completely ineffective the expected detection rate is 0%.

Treatment Condition: The proposed AntiGravity Shield countermeasure framework with all five coordinated detection modules in real-time. The framework transitions from periodic polling to kernel-event-driven monitoring (Linux `inotify`) and incorporates behavioural analysis, timestamp validation, canary-based deception and hash-chained log protection for improved detection.

For both conditions, the threat input is the Anti-Gravity autonomous agent, which is controlled and repeatable, and was developed and validated in the previous study (Orji et al., 2026). This means the only thing that changes between conditions is the independent variable, the monitoring architecture.

3.3.2 Modes of Attack

To evaluate the particular effect of anti-forensic techniques on detection results, the agent is tested in two operational modes:

Stealth Mode: The agent executes its anti-forensic operations (timestomping, secure file deletion, and log clearing) in a clean workspace with no concurrent background activity. This mode represents the idealised adversarial scenario and forms the primary experimental condition for the statistical hypothesis test (minimum 30 trials per condition).

Noisy Mode: The agent executes the same anti-forensic operations concurrently with a background workload that simulates routine legitimate file system activity (e.g., periodic file reads, writes, attribute changes, directory traversals, and log rotations). This introduces benign "noise" into the monitoring stream, testing the Shield's ability to discriminate between malicious and legitimate operations. Unlike stealth mode, the noisy-mode trials are not intended for primary statistical hypothesis testing; rather, they serve an exploratory and validation purpose. Critically, these noisy-mode sessions are run for significantly longer durations (e.g., several minutes) than the 10-second false-positive sanity check. During these extended sessions, the test harness actively records any WARNING or CRITICAL alerts generated by the background workload itself. This provides a more comprehensive empirical basis for assessing the Shield's false-positive rate under realistic operating conditions. A minimum of 3 noisy-mode trials are conducted.

3.3.3 Sample Size Justification

A minimum of 30 independent trials per primary condition (stealth mode) is specified. This threshold is selected for two reasons. First, it exceeds the minimum sample size commonly recommended for the application of the Central Limit Theorem, ensuring that the sampling distribution of the mean approximates normality regardless of the underlying data distribution (Field, 2018). Second, it directly addresses the methodological limitation noted by the AI-SS 2026 peer reviewers, who recommended a larger sample size to strengthen inferential claims. Additional trials are conducted in noisy mode (minimum 3) and for the false-positive baseline (minimum 1 extended monitoring session of 10+ seconds).

3.3.4 Human Baseline Comparison

A human baseline protocol models the same three anti-forensic techniques of timestomping, secure file deletion and log clearing, as manually executed by a human operator. The protocol emulates human latencies: command entry (1.5-6.0 seconds per command), viewing output, and order of operations. Such a comparison provides qualitative context to the experimental results, offering the scale of the speed difference between autonomous and human-speed anti-forensic operations. The human baseline is not a formal experimental condition, but helps to discuss why periodic monitoring fails. If a human operator can accomplish anti-forensic tasks in a single polling interval, then the vulnerability is not specific to AI but structural.

3.4 Data Collection Methods

3.4.1 Automated Test Harness

The complete experimental workflow for each trial is performed by an automated test harness ('test_harness.py') which removes human variability in the execution of experiments. The sequence that the harness performs is:

1. **Workspace Setup:** A clean isolated test workspace is created and the attack agent scripts copied on it.
2. **Shield Activation:** The AntiGravity Shield begins with all five modules active, monitoring the workspace directory. Canary files get deployed and the real time 'inotify' monitor begins looking for file system events.
3. **Stabilisation Pause:** 500 ms pause ensures the monitoring infrastructure is fully initialised before the attack begins.
4. **Implementation of Attack:** The Anti-Gravity agent is called as a sub-process in the monitored workspace. Attack start and end times are recorded with high precision timestamps ('time.perf_counter()').
5. **Event Propagation Stop:** The agent will wait 2.0 seconds to allow all 'inotify' kernel events to propagate through the detection pipeline.
6. **Shield Deactivation and Report Generation:** The Shield is halted, culminating in the last canary verification, external log integrity verifications, and the generation of a structured JSON report with all alerts, timing information, and aggregate statistics.
7. **Results Analysis:** The harness automatically classifies detections by type (timestomp, wiper signature, deletion, operation burst, canary tampering) and calculates detection completeness, latency, and number of alerts.

This automation guarantees that each trial is conducted in exactly the same manner, thus ensuring the internal validity of the experiment.

3.4.2 Quantitative Metrics

Four quantitative measures are recorded for each trial:

Detection latency – the time elapsed in milliseconds between the first anti-forensic action performed by the agent and the first alert generated by the Shield. This is measured with Python's 'time.perf_counter()' which is monotonic with sub-microsecond resolution and not affected by system clock updates.

Detection Completeness is the ratio of detected attack stages to the expected attack stages. Stealth mode detection will include four classifications: detection of timestomping, detection of wiper signatures, detection of file deletion and detection of operation burst. A completeness score of 100% indicated that all four categories were triggered in the trial.

False Positive Rate is the number of WARNING or CRITICAL alerts per unit of time during a quiescent operation without attack activity. A benign file workspace runs a preliminary false-positive test for at least 10 seconds without adversarial input. This short run verifies that the Shield does not generate spurious alerts under low background activity. A 10-second window cannot prove a global zero false-positive rate. The noisy-mode trials (Section 3.3.2) simulate legitimate user and system file operations for long periods to improve evaluation. During extended noisy-mode sessions, the test harness actively monitors and logs WARNING or CRITICAL alerts triggered by benign background activity, providing a more robust, but still limited, empirical assessment of the Shield's false-positive propensity. The goal is zero false-positives in the sanity check and extended noisy-mode trials.

Evidence Preservation The log integrity verification mechanism is used for assessment. The hash-chained audit log is verified after each trial. Should the chain be unbroken, the forensic record of the attack is retained even if the attacker attempted to sanitise logs inside the monitored workspace. A binary result (VERIFIED or COMPROMISED) is logged.

3.5 Statistical Analysis Plan

The primary statistical analysis compares the detection performance of the AntiGravity Shield (treatment) against the default periodic FIM (control) using the Mann-Whitney U test (Mann and Whitney, 1947). This non-parametric test is selected for the following reasons:

1. **No normality assumption:** Detection latency data are unlikely to be normally distributed because sub-second timing measurements tend to be right skewed. The Mann-Whitney U test is robust in this case as it does not assume normally distributed data (Field, 2018).
2. **Ordinal comparison:** The test can be used to determine whether the values of one distribution tend to be larger (or smaller) than the values of another, directly answering the research question whether the Shield achieves statistically superior detection performance.
3. **Small-to-moderate sample sizes:** The test is good with sample sizes of 30, the minimum specified for this study.

The convention in social science and information systems research sets the significance level at $p < 0.05$. If the null hypothesis (no difference between conditions) is rejected at this threshold, the alternative hypothesis that the integrated countermeasure framework provides significantly better detection performance is accepted.

Effect size is reported as the rank-biserial correlation (r) and provides a standardised estimate of the magnitude of difference between conditions that is independent of sample size. Effect sizes were interpreted using Cohen's (1988) guidelines for correlation coefficients: small ($r \approx 0.1$), medium ($r \approx 0.3$), and large ($r \approx 0.5$).

All quantitative measures are presented as descriptive statistics (mean, median, standard deviation, minimum and maximum). Box plots and bar charts are used to visualise results to enable comparison across conditions.

3.6 Threats to Validity

The following threats to validity are acknowledged:

Internal Validity: The controlled nature of the experiment ensures that the independent variable (monitoring architecture) is the primary variable affecting detection results. But a single attack agent with a fixed operational profile may not cover the full range of adversarial behaviour. The automated test harness provides consistent trial execution and removes experimenter bias.

External Validity: All experiments are performed in a closed simulation environment rather than a real enterprise network. The observed detection latencies and completeness rates may not directly translate to production environments with higher system load, network latency and more

complex file system activity. We recognise and discuss this limitation as a direction for future work.

Construct Validity: The four quantitative metrics (latency, completeness, false-positive rate, and evidence preservation) are chosen to provide a holistic view of the detection performance. However, they do not consider all aspects of defensive effectiveness, such as the quality of forensic attribution or the usability of alert information to incident responders.

3.7 Ethical Considerations

All experiments are run in isolated virtual environments not connected with production systems, live networks or real user data. The anti-forensic agent only works against mock files in sandboxed workspaces created by the test harness and destroyed after each run. There is at no point any personal, sensitive or third party data involved.

The research entails the development and use of offensive security tools (the Anti-Gravity agent). The agent is designed to operate only in the controlled experimental setting to minimise the ethical risks of dual-use research and is not released to the public. The capabilities of the agent are transparently documented in the prior publication (Orji et al., 2026) and this dissertation in accordance with the principle of responsible disclosure in security research (Rescorla, 2005).

A formal risk assessment has been undertaken in line with University of Salford requirements. The risk assessment indicates no significant risks to participants, data subjects or third parties. The research involves no human participants and no access to live systems.

